

Comment on Hybrid Graph–Machine Learning Framework for Accurate and Interpretable Band Gap Prediction

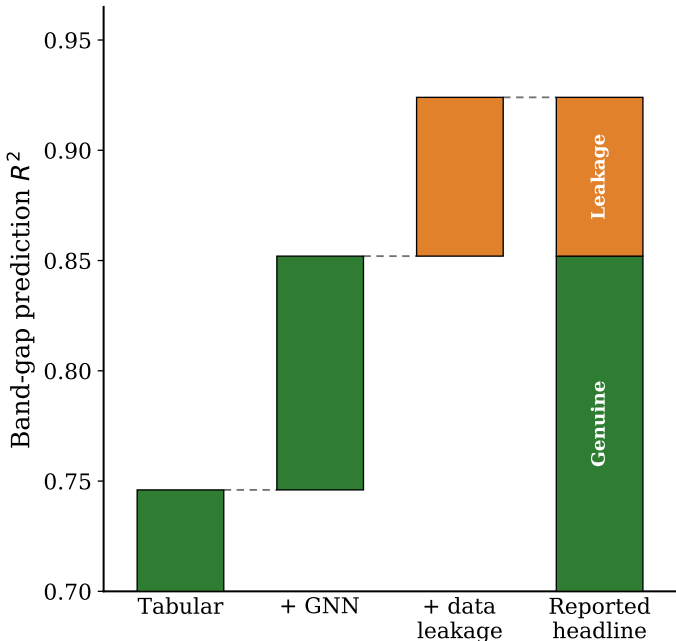
Shambhu Bhandari Sharma*

Department of Earth Sciences, University College London, London WC1E 6BT, United Kingdom

E-mail: shambhu.sharma.22@ucl.ac.uk

Abstract

Aydin et al. reported $R^2 = 0.921$ for band-gap prediction using hybrid graph neural networks. Reproduction shows this performance is inflated by data leakage from post-DFT proxies (`is_metal`, `is_gap_direct`) and embedding contamination. Controlled ablations reveal a genuine GNN gain of 0.08–0.11 in R^2 , while leakage contributes ~ 0.06 , lowering honest performance to 0.82–0.86 R^2 (MAE 0.28–0.30 eV).



Introduction

Materials-informatics models aim to circumvent the prohibitive cost of density-functional-theory (DFT) calculations.¹ The hybrid graph-neural-network and tabular framework proposed by Aydin et al. is methodologically well motivated.² Any such surrogate must, however, rely exclusively on inputs available *prior* to DFT simulation, namely chemical stoichiometry, crystal system, and structural topology.

The framework instead uses DFT-derived outputs and target-derived proxies as input features. This circular dependency constitutes data leakage and substantially inflates the reported performance. This Comment quantifies the resulting leakage and documents further flaws in feature attribution and evaluation protocol.

Data Leakage

Four distinct sources of data leakage affect the authors’ framework.

Tabular features. The clearest instance appears in the tabular feature set, which incorporates `is_metal`, `is_gap_direct`, `is_stable`, `is_magnetic`, and `num_magnetic_sites` as inputs. `is_metal` is the most severe of these, coinciding with `band_gap = 0` in essentially 100% of flagged materials. Because metals constitute 53.2% of the authors’ dataset (70,475 of 132,364 entries), this single feature reveals the target for over half the data, constituting fatal target leakage.

Pretrained MEGNet contamination. Embeddings were generated using the pretrained `Bandgap_MP_2018` MEGNet model,³ without accounting for evaluation materials already present in its training data. Matching Materials Project identifiers places 30.4% (40,186 of 132,364) of the evaluation materials within MEGNet’s 2018.6.1 dataset, of which 15.8% fall in the finite-gap subset on which the band-gap model was trained. The pretrained embedder cannot be regenerated without contamination, so this inflation cannot be isolated.

SchNet full-dataset leakage. The SchNet embedder⁴ was trained on the entire dataset without a holdout partition, so test embeddings were produced by a network that had already seen those structures.

CGCNN uncoordinated split. CGCNN’s⁵ training split was not coordinated with the final tabular split, producing train–test overlap.

Misattribution of Features

The authors describe `is_stable`, `is_gap_direct`, and `is_metal` as Magpie descriptors. This attribution is incorrect. Magpie computes stoichiometric descriptors from chemical formulae,⁶ whereas thermodynamic stability, direct-gap character, and metallicity derive from electronic-structure calculations, whether performed directly or retrieved from Materials Project property data. These variables are therefore post-DFT properties, not Magpie descriptors.

Methodological Discrepancies

The reported methodology diverges from the released code in several respects.

Cross-validation. The authors state that model performance was evaluated using 10-fold cross-validation. An audit of the codebase finds no such procedure. Models are evaluated on a single, static `train_test_split`. The only k -fold routine present is a 5-fold split used exclusively for target encoding, not the 10-fold model evaluation the manuscript describes.

Ensemble selection. Contrary to the authors’ claims, a three-model ensemble is not optimal. Testing on the 15% validation split (19,855 materials) shows that maximum accuracy requires five or six base models, never the three the paper reports.

Hyperparameter boundaries. The authors state that the Optuna⁷ search over the number of XGBoost estimators was restricted to 100–1000. The released configuration instead hardcodes `num_boost_round = 1336`, exceeding the stated upper bound.

Spin-orbit coupling. The authors describe a hybrid scheme combining DFTB values with literature estimates for missing spin-orbit-coupling parameters. The codebase instead applies the 2022 Jha & Heine DFTB set⁸ exclusively, setting missing elements to zero; no hybrid branch exists.

Ablation Analysis and Performance Decomposition

The authors’ pipeline was first reproduced on their published dataset (DOI 10.5281/zenodo.18481208) to establish a baseline for the ablations. The reproduced stacking R^2 recovers CGCNN exactly at 0.909, MEGNet at 0.924 against a reported 0.921, and SchNet at 0.883 (Figure 1, solid bars). This agreement confirms a faithful reimplementaion.

With that baseline established, the ablations sequentially remove tabular and embedding leakage, refitting all six base regressors and the RidgeCV ensemble on the same 70/15/15 split at each stage. The tabular ablation excludes the post-DFT proxy features `is_metal`, `is_gap_direct`, `is_stable`, `is_magnetic`, and `num_magnetic_sites`. Graph embeddings are then regenerated with the held-out test structures excluded from GNN training. This embedding ablation is restricted to SchNet and CGCNN; MEGNet is excluded because its pretrained 2018 embeddings cannot be regenerated without test-set contamination. SHAP analysis confirms the intended effect, with `predicted_band_gap` dominating in all three leakage-free graph arms and `MagpieData maximum Electronegativity` leading in the tabular baseline.

The leakage-free tabular baseline reaches 0.746 R^2 , a value the original paper never reports despite its necessity for any fair assessment of the GNN contribution. Against this baseline, CGCNN stacking achieves the largest honest gain, 0.110 R^2 , reaching 0.856 R^2 , while SchNet reaches 0.825 R^2 , a gain of 0.078 R^2 . MEGNet reaches 0.860 R^2 , only marginally above CGCNN despite retaining test-set contamination. This narrow margin indicates that contamination did not meaningfully favour MEGNet, likely because its pretrained embeddings derive from an older and smaller 2018 training set. Measured against the authors’ reported values, leakage accounts for between 0.046 and 0.061 R^2 across the three architectures, raising the reported MAE of 0.191 eV to 0.275–0.302 eV.

Conclusion

The hybrid framework is a genuinely valuable architectural concept, delivering a measurable lift over the tabular baseline. However, the reported accuracy incorporates several distinct leakage sources, namely target-derived tabular proxies, in-sample graph embeddings, and pretrained-embedder contamination of the test set. Controlled ablations show CGCNN to

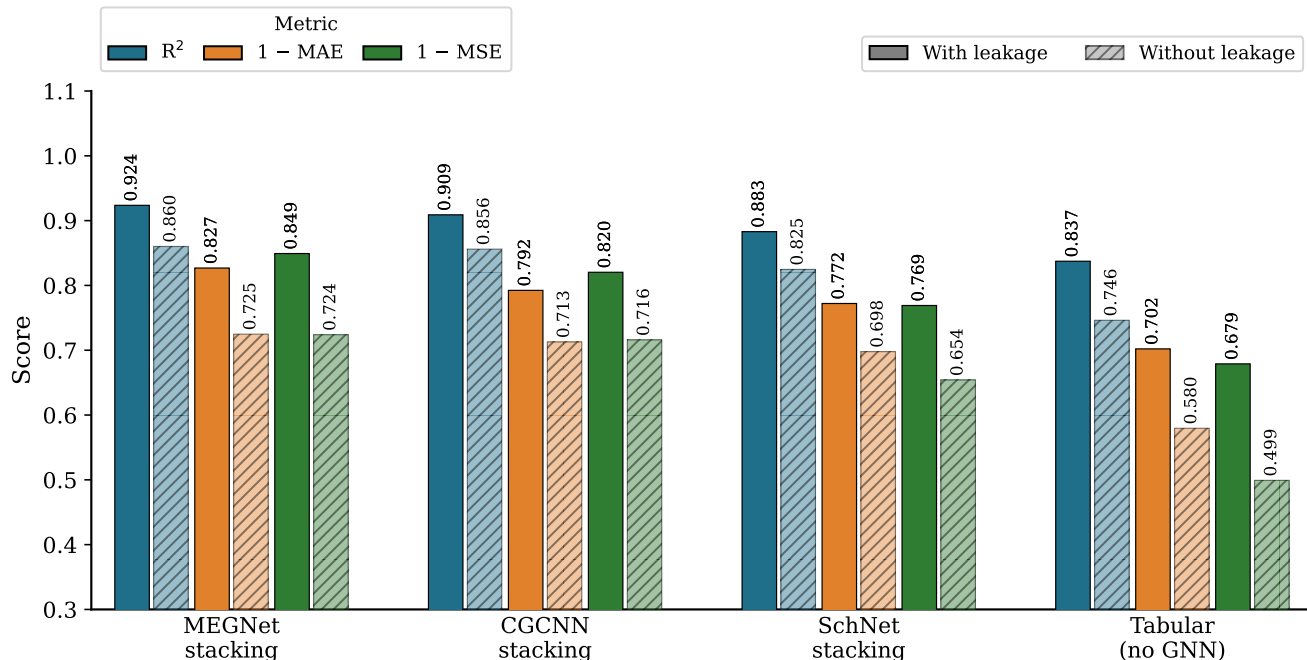


Figure 1: Performance decomposition on the held-out test set. Solid bars reproduce the authors’ published stacking ensemble (“with leakage”); hatched bars show the same ensemble refitted after removing the post-DFT proxy features, with graph embeddings additionally regenerated leakage-free for CGCNN and SchNet (“without leakage”). Metrics are R^2 , $1 - \text{MAE}$, and $1 - \text{MSE}$. Performance falls uniformly, and the drop is largest for the tabular model, which has no embeddings to carry the signal. MEGNet’s pretrained embedder cannot be regenerated (discussed above), so its hatched bars retain test-set contamination and remain an upper bound.

be the strongest honest performer among the studied architectures. This overturns the original ranking, in which MEGNet’s apparent advantage rests on test-set exposure rather than demonstrated architectural merit. Combined with the misattributed feature provenance and the divergences between the reported and released methodology, these findings highlight the value of careful code review and explicit feature-validity criteria in materials informatics.

Methods

Pipelines were reconstructed from the authors’ codebase and applied to their published dataset (DOI 10.5281/zenodo.18481208; $N = 132,364$). Models were trained on a 70/15/15 split (92,654/19,855/19,855 entries; `random_state = 42`), stratified on binned band gap, with all six base regressors and the RidgeCV ensemble retrained from scratch for each ablation. Retractable embeddings (CGCNN, SchNet) were regenerated with the held-out test structures excluded from GNN training, using the training epochs and learning rates of the upstream configuration. MEGNet contamination was quantified by intersecting Materials Project identifiers with the 2018.6.1 training set, and feature importances were computed

with SHAP.

Data and Software Availability

All code, models, Jupyter notebooks, and processed data required to reproduce the audit and findings of this study are openly available on GitHub at <https://github.com/shamhubhandari/Hybrid-GNN>.

Acknowledgement

The author thanks the authors of the original study for making their dataset and codebase openly available, without which this analysis would not have been possible, and acknowledges the use of the UCL High Performance Computing Facility.

References

- (1) Kohn, W.; Sham, L. J. Self-consistent equations including exchange and correlation effects. *Physical review* **1965**, *140*, A1133.
- (2) Aydın, A.; Eryılmaz, Ü. K.; Alkan, O. B.; Kocagöz, P.; Ekinci, F.; Guzel, M. S. Hybrid Graph-Machine Learning Framework for Accurate and Interpretable Band Gap Prediction. *Journal of Chemical Information and Modeling* **2026**, *66*, 3787–3802.
- (3) Chen, C.; Ye, W.; Zuo, Y.; Zheng, C.; Ong, S. P. Graph Networks as a Universal Machine Learning Framework for Molecules and Crystals. *Chem. Mater.* **2019**, *31*, 3564–3572.
- (4) Schütt, K.; Kindermans, P.-J.; Sauceda Felix, H. E.; Chmiela, S.; Tkatchenko, A.; Müller, K.-R. Schnet: A continuous-filter convolutional neural network for modeling quantum interactions. *Advances in neural information processing systems* **2017**, *30*.
- (5) Xie, T.; Grossman, J. C. Crystal graph convolutional neural networks for an accurate and interpretable prediction of material properties. *Physical review letters* **2018**, *120*, 145301.
- (6) Ward, L.; Agrawal, A.; Choudhary, A.; Wolverton, C. A general-purpose machine learning framework for predicting properties of inorganic materials. *npj Computational Materials* **2016**, *2*, 16028.
- (7) Akiba, T.; Sano, S.; Yanase, T.; Ohta, T.; Koyama, M. Optuna: A next-generation hyperparameter optimization framework. Proceedings of the 25th ACM SIGKDD international conference on knowledge discovery & data mining. 2019; pp 2623–2631.
- (8) Jha, G.; Heine, T. DFTB parameters for the periodic table: Part III, spin-orbit coupling. *Journal of chemical theory and computation* **2022**, *18*, 4472–4481.